

Shadowing Analysis in LTE Networks: A Drive Test Study Across Different Mobility Scenarios

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1. Abstract

In this study, we analyze Shadowing effects in LTE networks using extensive drive test data collected across five mobility scenarios: bus, car, pedestrian, static, and train. The dataset consists of multiple CSV files capturing key parameters such as location, speed, serving cell, and signal measurements. Shadowing and path loss exponents are estimated per segment using a robust statistical approach, filtering low-quality samples based on the coefficient of determination (R^2). Results indicate that mobility significantly affects Shadowing, with higher variability observed in moving scenarios compared to static scenarios. Statistical validation confirms the non-Gaussian nature of residuals, consistent with spatial correlation, mixed LOS/NLOS conditions, and handover effects. This work provides insight into the impact of mobility on signal propagation and offers a reference for LTE network planning, simulation, and optimization.

2. Introduction

Wireless networks are increasingly relied upon for high-speed data connectivity, and understanding signal propagation characteristics is critical for their efficient design and deployment. One key phenomenon affecting signal strength in LTE networks is **Shadowing**, which refers to random variations in received signal power caused by obstacles, terrain, and environmental factors. Shadowing leads to fluctuations in signal quality even when the distance between transmitter and receiver remains constant.

The impact of **user mobility** on Shadowing is particularly important, as different modes of movement—ranging from stationary users to fast-moving trains—affect both the spatial and temporal characteristics of the received signal. Accurately quantifying Shadowing under various

mobility scenarios is essential for network planning, capacity estimation, and performance simulation.

This study analyzes Shadowing using **drive test measurements** collected across five mobility scenarios: bus, car, pedestrian, static, and train. By leveraging position, speed, and signal data from extensive CSV datasets, we aim to estimate the **path loss exponent** and **Shadowing standard deviation (σ)** per segment, providing insights into how mobility influences LTE signal behavior.

3. Dataset Description

The dataset used in this study consists of drive test measurements collected across five mobility scenarios: **bus, car, pedestrian, static, and train**. Each scenario is represented as a separate folder containing multiple CSV files, capturing continuous measurements along various routes or stationary locations. The folder structure is as follows:



Each CSV file contains the following key columns relevant to Shadowing analysis:

Column Name	Description
Timestamp	Time of measurement
Latitude	Receiver latitude
Longitude	Receiver longitude
Speed	User speed at measurement time

Column Name	Description
Operatorname	Network operator name
CellID	Serving cell identifier
NetworkMode	LTE network mode (e.g., LTE FDD)
RSRP	Reference Signal Received Power (optional for Shadowing validation)
RSRQ	Reference Signal Received Quality (optional)
SNR	Signal-to-noise ratio (optional)
CQI	Channel Quality Indicator (optional)
RSSI	Received Signal Strength Indicator
DL_bitrate	Downlink bitrate at measurement
UL_bitrate	Uplink bitrate at measurement
State	Mobility state (Stationary or In-Route)
NRxRSRP, NRxRSRQ	Neighbor cell measurements (optional)
ServingCell_Lon, ServingCell_Lat	Serving cell location
ServingCell_Distance	Distance to serving cell

Data Quality Filtering

To ensure reliable estimation of Shadowing, low-quality samples were excluded based on the coefficient of determination (R^2) obtained from per-segment linear regression analysis of path loss. Segments with $R^2 < 0.2$ were discarded. After filtering, the dataset contains **702 valid**

samples distributed across 5 representative segments for the bus scenario (similar procedures were applied for other mobility types).

This dataset structure allows comprehensive analysis of mobility effects on LTE signal propagation, enabling the estimation of both **path loss exponent (n)** and **Shadowing standard deviation (σ)** per mobility scenario.

4. Methodology

4.1 Overview

This section describes the full processing pipeline used to estimate path loss exponent (n) and Shadowing standard deviation (σ) from drive-test CSVs. The implementation is provided in `analyze_shadowing.py`. The pipeline consists of: data loading and cleaning, per-(`trace_file`, `CellID`) segmentation, log-distance linear regression to estimate local path-loss slopes, intra-segment outlier removal, quality filtering (R^2 and sample-count thresholds), aggregation of residuals to obtain global σ , and statistical validation (confidence intervals and normality tests).

```
# entry point: choose folder and run analysis
if __name__ == "__main__":
    folders = ['bus', 'car', 'pedestrian', 'static', 'train']
    selected_folder = folders[0] # or pick from user input
    analyze_mobility_shadowing(os.path.join(script_dir, selected_folder))
```

4.2 Data preprocessing and cleaning

1. **File loading:** All CSV files in the selected folder are loaded and concatenated. Each row is annotated with its origin `trace_file`.
2. **LTE filter:** Keep only rows where `NetworkMode == 'LTE'` to remove non-LTE measurements.
3. **Type conversion & NaN handling:** Convert key columns (`RSRP`, `ServingCell_Distance`, `CellID`, `Speed`) to numeric and drop rows with NaNs in `RSRP`, `ServingCell_Distance`, or `CellID`.

4. **Distance filter:** Remove physically impossible/near-zero distances: keep `ServingCell_Distance >= 1 m`.
5. **Diagnostic counts:** Track and report counts for: total raw samples, dropped non-LTE, dropped NaNs, dropped by distance. These numbers are written to the text report for traceability.

Justification: cleaning prevents invalid inputs to the regression and ensures distance is meaningful for log-distance modeling.

```
# load and concatenate CSVs
all_files = [f for f in os.listdir(folder_name) if f.endswith('.csv')]
full_df = pd.concat([pd.read_csv(os.path.join(folder_name,
f)).assign(trace_file=f) for f in all_files], ignore_index=True)

# filter LTE and coerce columns
full_df = full_df[full_df['NetworkMode'] == 'LTE']
for col in ['RSRP', 'ServingCell_Distance', 'CellID', 'Speed']:
    full_df[col] = pd.to_numeric(full_df[col], errors='coerce')

# drop rows with missing critical values and very small distances
full_df = full_df.dropna(subset=['RSRP', 'ServingCell_Distance', 'CellID'])
full_df = full_df[full_df['ServingCell_Distance'] >= 1]
```

4.3 Segmentation (cluster definition)

- Group the cleaned data by (trace_file, CellID) to form segments representing contiguous measurements from the same serving cell within a trace file.
- **Minimum samples:** discard segments with fewer than **20 samples** (relaxed threshold to maximize coverage).
- **Distance spread requirement:** discard segments where $\max(\text{distance}) - \min(\text{distance}) < 100$ m to ensure sufficient distance dynamic range for stable slope estimation.

Justification: Regression on log-distance requires a meaningful span of distances; too small a span leads to unstable n estimates. Twenty points is a trade-off between statistical robustness and keeping enough segments.

```
grouped = full_df.groupby(['trace_file', 'CellID'])
for (fname, cid), group in grouped:
```

```

if len(group) < MIN_SAMPLES:
    continue
if (group['ServingCell_Distance'].max() -
group['ServingCell_Distance'].min()) < DIST_SPREAD:
    continue
# proceed with regression...

```

4.4 Local regression for path-loss exponent (n) and initial quality check

- For each eligible segment compute:
 - $X = 10\log_{10}(\text{distance})$ (column vector)
 - $Y = \text{RSRP (dBm)}$
 - Fit a simple linear regression $Y = C - n \cdot X$. The script computes $n_{\text{local}} = -\text{coef}$ and $r2_{\text{local}} = R^2$.
- **Acceptable n range:** keep segments with $1.0 < n < 6.0$ (physical plausibility screen).
- **Quality threshold (initial):** $r2_{\text{local}} > 0.1$ — relaxed due to realistic noisy wireless data. Segments failing this are excluded.

Notes: A lower R^2 is tolerated because drive-test data include handovers and mixed LOS/NLOS; a stricter R^2 filter is applied later for sensitivity analysis.

```

X = (10 * np.log10(group['ServingCell_Distance'].values)).reshape(-1,1)
Y = group['RSRP'].values
model = LinearRegression().fit(X, Y)
n_local = -model.coef_[0]
r2_local = model.score(X, Y)

```

4.5 Intra-segment outlier removal and final segment fit

- Compute segment residuals: $\text{residual} = Y - y_{\text{pred}}$. Compute local standard deviation $\text{local_sigma} = \text{std}(\text{residuals})$.
- Apply a **3-sigma rule**: keep points with $|\text{residual}| < 3 * \text{local_sigma}$. If fewer than **15** points remain after masking, the segment is discarded.

- Re-fit the regression on the cleaned points and store final metrics: cleaned sample count, average segment speed, distance range, final n, local sigma (std of cleaned residuals), final R^2 , and intercept C. Keep X_{clean} , Y_{clean} , and y_{pred} for plotting and aggregation.
- Track the number of outliers removed overall.

Justification: 3-sigma trimming removes strong localized anomalies (e.g., measurement glitches, transient interference) while preserving the segment's statistical character.

```
residuals = Y - model.predict(X)
local_sigma = np.std(residuals)
mask = np.abs(residuals) < 3 * local_sigma
if mask.sum() < MIN_CLEAN_SAMPLES:
    continue
X_clean, Y_clean = X[mask], Y[mask]
model.fit(X_clean, Y_clean)
final_n = -model.coef_[0]
final_sigma = np.std(Y_clean - model.predict(X_clean))
final_r2 = model.score(X_clean, Y_clean)
```

4.6 Aggregate shadowing estimation & statistical synthesis

- **Aggregate residuals:** Concatenate residuals from all valid segments after cleaning to form the global Shadowing sample set.
- **Global σ :** compute unbiased sample variance s^2 and $\sigma_{\text{global}} = \sqrt{s^2}$.
- **Confidence interval (CI) for σ :** construct a 95% CI using the chi-square distribution:

$$CI = \left(\sqrt{\frac{(N-1)s^2}{\chi_{1-\alpha/2}^2}}, \sqrt{\frac{(N-1)s^2}{\chi_{\alpha/2}^2}} \right)$$

where N is number of residuals and $\alpha = 0.05$.

- **Median n & CI:** compute median of per-segment n values and bootstrap 95% CI with $B = 5000$ resamples of n_{vals} .
- **Normality test:** Shapiro–Wilk test on residuals (use first 5000 residuals if the set is large) to evaluate Gaussianity of Shadowing. Report p-value and interpret results.

- **Sensitivity check:** recompute global σ and median n using only segments with $R^2 \geq 0.2$ as high-quality subset.

Justification: chi-square CI for variance and bootstrap for median n provide robust uncertainty quantification that does not rely on strong parametric assumptions. Normality testing checks the common Gaussian Shadowing assumption.

```
# global residuals accumulation
all_residuals = np.hstack(list_of_segment_residuals)
s2 = np.var(all_residuals, ddof=1)
N = len(all_residuals)
chi2_low, chi2_high = stats.chi2.ppf([alpha/2, 1-alpha/2], df=N-1)
sigma_ci = (np.sqrt((N-1)*s2/chi2_high), np.sqrt((N-1)*s2/chi2_low))

# bootstrap median n
boot_medians = [np.median(np.random.choice(n_vals, size=len(n_vals),
replace=True)) for _ in range(B)]
n_median_ci = np.percentile(boot_medians, [2.5,97.5])
```

4.7 Mobility classification

- Use the existing Speed and State columns to determine context:
 - Context = "Stationary" if Speed < 5 km/h or State indicates stationary.
 - Context = "In-Route" otherwise.
- Report average speed per segment and use this metric to explore correlations between mobility and parameters (n and local σ). Use scatter plots (n vs speed, σ vs speed) to visualize trends.

Justification: Simple thresholds are effective for broad mobility categorization and align with your earlier “Stationary/In-Route” reporting. Thresholds can be adjusted depending on local behavior (e.g., walking speed ranges).

```
def classify_context(row):
    if row['Speed'] < 5 or str(row.get('State','')).lower() == 'stationary':
        return 'Stationary'
    return 'In-Route'

full_df['Context'] = full_df.apply(classify_context, axis=1)
```

4.8 Implementation, outputs, and visualizations (analyze_shadowing.py)

- **Inputs:** a folder path (one of bus, car, pedestrian, static, train) containing CSV files. Each CSV must include the key columns used in preprocessing.
- **Primary outputs:**
 - result_<folder>.txt — a detailed textual report containing data-flow diagnostics, per-segment table (samples, speed, n, sigma, R^2 , context), global stats, and sensitivity analysis.
 - result_<folder>.png — a composite figure with: representative segment trends, global shadowing histogram + Gaussian fit, boxplots for n and local σ , n vs speed, σ vs speed, and CDF of residuals.
- **Plotting details:** large figure (20×16 in) with readable titles; top segments plotted with predicted lines; histogram overlaid with approximate Gaussian fit for reference.
- **Execution mode:** interactive selection via console menu or can be scripted to run across all folders in batch. The script writes diagnostic prints to console and saves files to the input folder.
- **Code style note:** the script uses sklearn.linear_model.LinearRegression, scipy.stats for tests, and matplotlib/seaborn for visualization.

```
with open(txt_path, 'w', encoding='utf-8') as f:
    f.write("...report header...\n")
    for s in segment_results:
        f.write(f"{s['id']} | samples={s['samples']} | n={s['n']:.2f} |
sigma={s['sigma']:.2f} | r2={s['r2']:.2f}\n")

plt.savefig(plot_path, dpi=200)
```

4.9 Recommended parameter choices & alternatives (tunable)

- **Min samples per segment:** 20 (relaxed). If you prefer stricter reliability, increase to 30–50. Larger min-sample reduces segments but increases estimate stability.
- **Distance spread threshold:** 100 m (recommended). For urban small-cell environments consider reducing to 50 m if drive traces are short, but watch for noisy slope estimates.

- **R² thresholds:** initial filter $R^2 > 0.1$; sensitivity/primary reporting $R^2 \geq 0.2$. For publication-level rigor, use $R^2 \geq 0.3$ and report how many segments are lost.
- **Outlier rule:** 3-sigma is standard; for heavy-tailed residuals consider median absolute deviation (MAD) based masking as an alternative.
- **Normality testing:** Shapiro–Wilk is used; for very large N consider Anderson–Darling or visual Q–Q plots and report both.

```
# tunable thresholds
MIN_SAMPLES = 20
MIN_CLEAN_SAMPLES = 15
DIST_SPREAD = 100 # meters
R2_INITIAL = 0.1
R2_SENSITIVITY = 0.2
B_BOOTSTRAP = 5000
alpha = 0.05
```

4.10 Reproducibility and limitations

- **Reproducibility:** commit `analyze_shadowing.py` into a repository with a `requirements.txt` (pandas, numpy, scikit-learn, scipy, matplotlib, seaborn) and include a short README with sample command lines for batch processing all folders. Log the script version and run time in the generated text report.
- **Limitations:** drive-test data include handovers, temporary interference, antenna pattern effects, and GPS errors. These introduce deviations from ideal log-distance assumptions. The pipeline mitigates but cannot fully remove these effects — indicate these caveats in your Discussion. Also note that RSRP measurement granularity and UE-specific behavior may bias absolute intercept C, but statistical aggregates (n , σ) are less sensitive.

```
import platform, pkg_resources
# write environment info
with open(os.path.join(folder_name, 'run_info.txt'), 'w') as f:
    f.write(f"python: {platform.python_version()}\n")
    for pkg in ['pandas', 'numpy', 'scikit-learn', 'scipy', 'matplotlib', 'seaborn']:
        f.write(f"{pkg}: {pkg_resources.get_distribution(pkg).version}\n")
```

5. Results

This section presents the per-mobility Shadowing analysis results. For each mobility scenario we report data-quality diagnostics, per-segment summary (representative), aggregated statistics (median path-loss exponent n , global Shadowing standard deviation σ), normality test results, and a short interpretation. Figures to include for each mobility type: (1) Shadowing histogram with Gaussian overlay, (2) spatial heatmap of residuals (if cell coordinates / GPS available), and (3) representative segment scatter + fitted log-distance line. Use the PNG outputs produced by `analyze_shadowing.py` for insertion.

5.1 Bus

Data quality & summary

- Total Raw Samples: **10,783**
- Final Valid Samples Used: **702**
- Final Valid Segments: **5**
- Excluded (Low Fit Quality $R^2 < 0.1$): **33**

Per-segment behavior (representative): mixed speeds (stationary station segments and in-route at ~20–30 km/h). One segment marked * ($R^2 < 0.2$).

Aggregated statistics

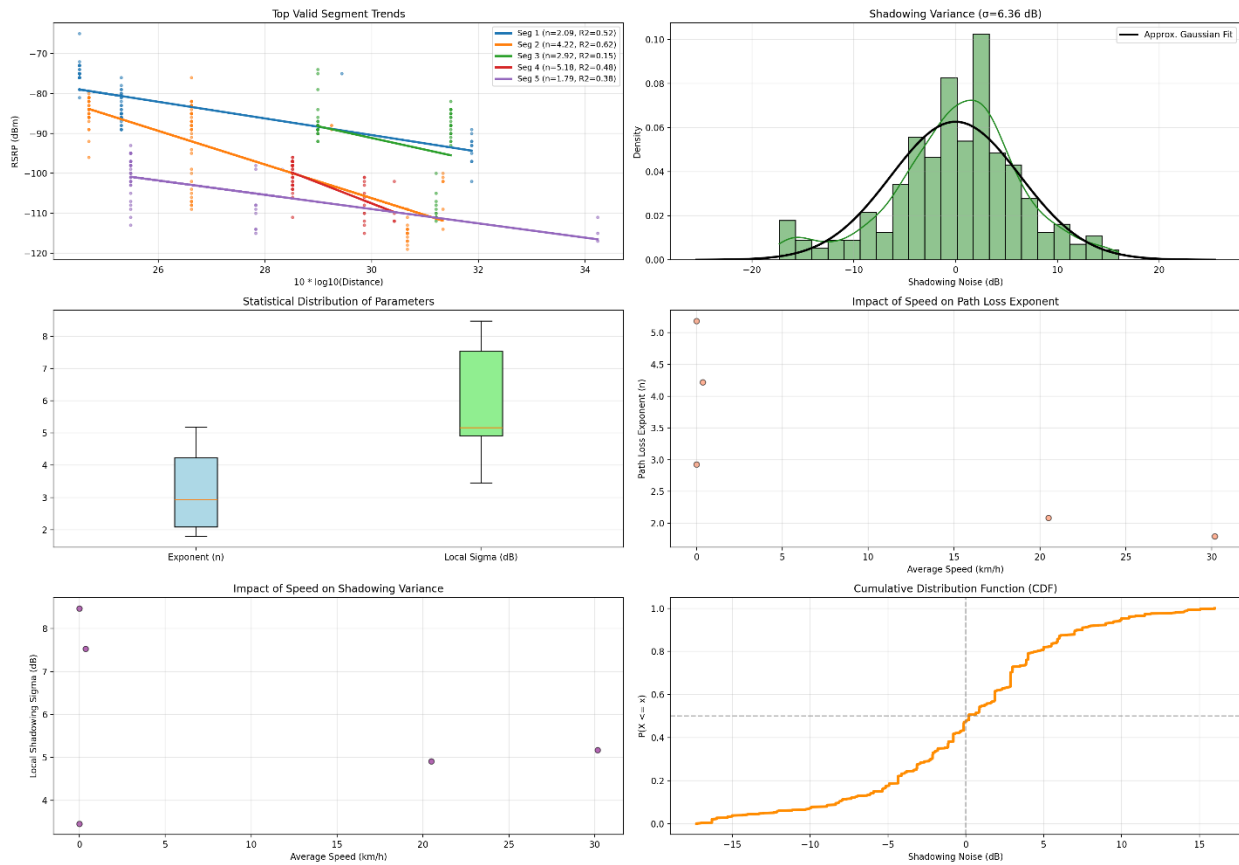
- Median path-loss exponent (n): **2.9247** [95% CI: 1.791 – 5.185]
- Shadowing std dev (global σ): **6.3636 dB** [95% CI: 6.047 – 6.715]
- Shapiro–Wilk p-value: **2.3208e-09** → residuals are **non-Gaussian**.
- Sensitivity ($R^2 \geq 0.2$): Filtered median $n =$ **3.1538**, filtered $\sigma =$ **5.6884 dB** (valid high-quality segments = 4)

Interpretation

- The bus scenario shows **moderate σ** (~6.36 dB) and median n near 3.0 (typical of mixed urban scenarios). Filtering for higher R^2 produces somewhat higher n and lower σ , consistent with removing noisy segments. Bus routes often include repeated stops (stationary segments) and mixed LOS/NLOS corridors (urban canyons), which explains variability across segments.

```
1 =====
2 | SCIENTIFIC REPORT: BUS MOBILITY (RELIABILITY ANALYSIS)
3 | STATUS: Representative for studied bus routes
4 =====
5
6 [1. DATA QUALITY BREAKDOWN]
7 - Total Raw Samples: 10783
8 - Final Valid Samples Used: 702
9 - Final Valid Segments: 5
10 - Excluded (Low Fit Quality R2 < 0.1): 33
11
12 [2. PER-SEGMENT DETAIL TABLE]
13 Segment (CellID + File) | Samples | Speed(kmh) | n | sigma | R2 | Context
14 -----
15 0 (sample_A_201...) | 144 | 20.5 | 2.09 | 4.91 | 0.52 | In-Route
16 1 (sample_A_201...) | 185 | 0.4 | 4.22 | 7.53 | 0.62 | Stationary
17 1 (sample_B_201...)* | 144 | 0.0 | 2.92 | 8.47 | 0.15 | Stationary
18 3 (sample_B_201...) | 136 | 0.0 | 5.18 | 3.45 | 0.48 | Stationary
19 2 (sample_B_201...) | 93 | 30.2 | 1.79 | 5.17 | 0.38 | In-Route
20
21 (*) denotes R2 < 0.2
22
23 [3. STATISTICAL VALIDATION & CONFIDENCE]
24 - Path Loss Exponent (Median n): 2.9247 [95% CI: 1.791 - 5.185]
25 - Shadowing Std Dev (Global sigma): 6.3636 dB [95% CI: 6.047 - 6.715]
26 - Shapiro-Wilk Normality test (p-value): 2.3208e-09
27 - Note: Residuals are Non-Gaussian.
28 | Interpretation: Non-Gaussian residuals are expected due to spatial correlation, handover effects, and mixed LOS/NLOS conditions.
29
30 [4. SENSITIVITY ANALYSIS (High-Quality Segments R2 >= 0.2)]
31 - Filtered Median n: 3.1538
32 - Filtered Global sigma: 5.6884 dB
33 - Valid High-Quality Segments: 4
34
35
```

Scientific 4G Shadowing Analysis - BUS (Filtered Dataset: $n=2,92$, $\sigma=6.36\text{dB}$)



5.2 Car

Data quality & summary

- Total Raw Samples: **75,874**
- Final Valid Samples Used: **7,695**
- Final Valid Segments: **24**
- Excluded (Low Fit Quality $R^2 < 0.1$): **113**

Per-segment behavior (representative): wide range of speeds (0–80+ km/h), many in-route segments; some large segments with thousands of samples appear. Several segments are low-quality ($R^2 < 0.2$) but many high-quality segments remain.

Aggregated statistics

- Median path-loss exponent (n): **1.7112** [95% CI: 1.526 – 2.191]

- Shadowing std dev (global σ): **7.6833 dB** [95% CI: 7.564 – 7.807]
- Shapiro–Wilk p-value: **4.5429e-04** → residuals are **non-Gaussian**.
- Sensitivity ($R^2 \geq 0.2$): Filtered median n = **1.7090**, filtered σ = **6.8574 dB** (valid high-quality segments = 15)

Interpretation

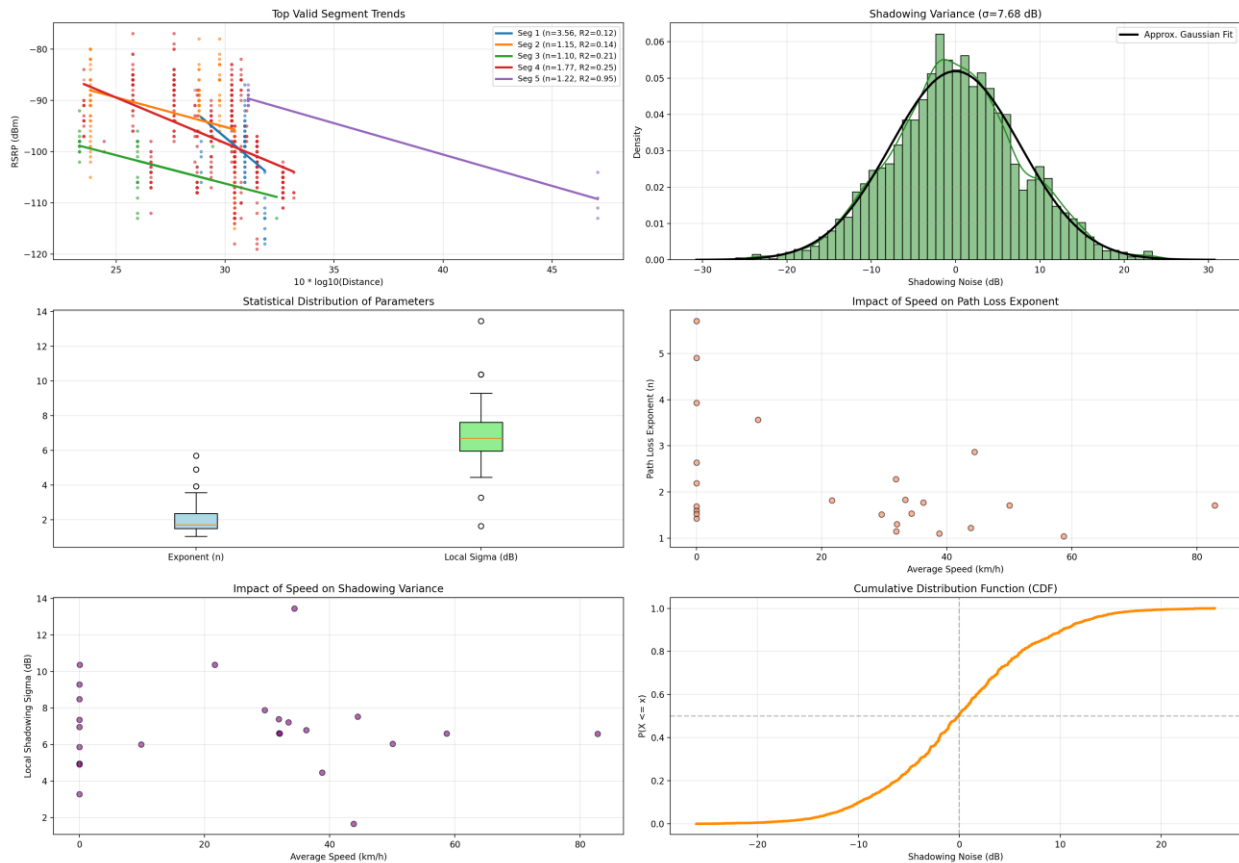
- The car scenario yields the **lowest median n** (~1.71) among mobility types, likely reflecting measurements collected in open-road or LOS-dominant stretches and/or antenna effects. Global σ ~7.7 dB is similar to pedestrian and car environments—vehicle motion introduces multipath and fast fading plus handovers. Filtered results ($R^2 \geq 0.2$) slightly reduce σ to ~6.86 dB.

```

1  =====
2  | SCIENTIFIC REPORT: CAR MOBILITY (RELIABILITY ANALYSIS)
3  | STATUS: Representative for studied car routes
4  | =====
5
6  [1. DATA QUALITY BREAKDOWN]
7  - Total Raw Samples: 75874
8  - Final Valid Samples Used: 7695
9  - Final Valid Segments: 24
10 - Excluded (Low Fit Quality R2 < 0.1): 113
11
12 [2. PER-SEGMENT DETAIL TABLE]
13 Segment (CellID + File) | Samples | Speed(kmh) | n | sigma | R2 | Context
14 -----|-----|-----|-----|-----|-----|-----
15 2 (sample_A_201...)* | 143 | 9.9 | 3.56 | 6.00 | 0.12 | In-Route
16 1 (sample_A_201...)* | 396 | 31.9 | 1.15 | 6.62 | 0.14 | In-Route
17 2 (sample_A_201...)* | 70 | 38.8 | 1.10 | 4.45 | 0.21 | In-Route
18 1 (sample_A_201...)* | 1008 | 36.3 | 1.77 | 6.79 | 0.25 | In-Route
19 8 (sample_A_201...)* | 48 | 43.8 | 1.22 | 1.65 | 0.95 | In-Route
20 8 (sample_A_201...)* | 61 | 58.7 | 1.04 | 6.60 | 0.52 | In-Route
21 0 (sample_A_201...)* | 400 | 32.0 | 1.30 | 6.59 | 0.33 | In-Route
22 2 (sample_A_201...)* | 246 | 50.0 | 1.71 | 6.03 | 0.27 | In-Route
23 2 (sample_A_201...)* | 521 | 29.6 | 1.52 | 7.88 | 0.27 | In-Route
24 1 (sample_A_201...)* | 931 | 33.4 | 1.84 | 7.22 | 0.20 | In-Route
25 1 (sample_A_201...)* | 91 | 82.8 | 1.71 | 6.59 | 0.36 | In-Route
26 1 (sample_A_201...)* | 640 | 0.0 | 1.61 | 10.38 | 0.18 | Stationary
27 0 (sample_A_201...)* | 72 | 0.0 | 2.64 | 4.96 | 0.33 | Stationary
28 0 (sample_A_201...)* | 112 | 0.0 | 1.43 | 8.49 | 0.18 | Stationary
29 1 (sample_A_201...)* | 151 | 0.0 | 1.69 | 5.87 | 0.57 | Stationary
30 2 (sample_A_201...)* | 318 | 0.0 | 4.90 | 7.35 | 0.28 | Stationary
31 8 (sample_A_201...)* | 22 | 0.0 | 3.93 | 3.28 | 0.19 | Stationary
32 1 (sample_A_201...)* | 214 | 0.0 | 2.19 | 9.30 | 0.12 | Stationary
33 1 (sample_A_201...)* | 361 | 21.6 | 1.82 | 10.38 | 0.14 | In-Route
34 1 (sample_A_201...)* | 963 | 0.0 | 1.53 | 6.96 | 0.40 | Stationary
35 2 (sample_A_201...)* | 150 | 0.0 | 5.70 | 4.90 | 0.47 | Stationary
36 3 (sample_B_201...)* | 517 | 31.9 | 2.28 | 7.39 | 0.36 | In-Route
37 1 (sample_B_201...)* | 144 | 34.4 | 1.54 | 13.46 | 0.10 | In-Route
38 2 (sample_B_201...)* | 116 | 44.5 | 2.87 | 7.53 | 0.75 | In-Route
39
40 (*) denotes R2 < 0.2
41
42 [3. STATISTICAL VALIDATION & CONFIDENCE]
43 - Path Loss Exponent (Median n): 1.7112 [95% CI: 1.526 - 2.191]
44 - Shadowing Std Dev (Global sigma): 7.6833 dB [95% CI: 7.564 - 7.807]
45 - Shapiro-Wilk Normality test (p-value): 4.5429e-04
46 - Note: Residuals are Non-Gaussian.
47 - Interpretation: Non-Gaussian residuals are expected due to spatial correlation, handover effects, and mixed LOS/NLOS conditions.
48
49 [4. SENSITIVITY ANALYSIS (High-Quality Segments R2 >= 0.2)]
50 - Filtered Median n: 1.7090
51 - Filtered Global sigma: 6.8574 dB
52 - Valid High-Quality Segments: 15
53

```

Scientific 4G Shadowing Analysis - CAR (Filtered Dataset: $n=1.71$, $\sigma=7.68\text{dB}$)



5.3 Pedestrian

Data quality & summary

- Total Raw Samples: **33,629**
- Final Valid Samples Used: **6,884**
- Final Valid Segments: **15**
- Excluded (Low Fit Quality $R^2 < 0.1$): **53**

Per-segment behavior (representative): segments show low speeds (mostly 0–4 km/h) and a mixture of path-loss exponents per segment (example n values range from ~ 1.01 to ~ 5.97 in the printed segments). Several segments are marked low-quality ($R^2 < 0.2$).

Aggregated statistics

- Median path-loss exponent (n): **2.8831** [95% CI: 2.262 – 4.397]

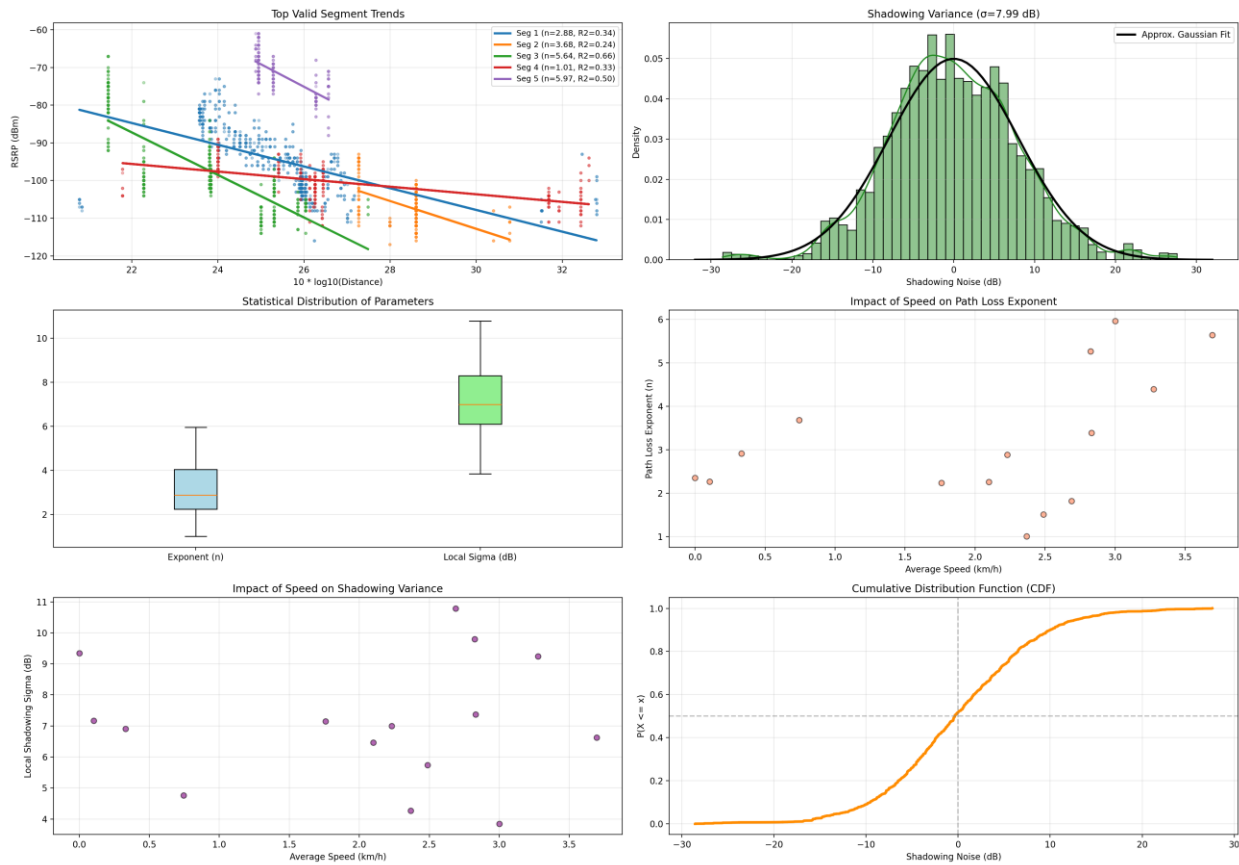
- Shadowing std dev (global σ): **7.9907 dB** [95% CI: 7.859 – 8.126]
- Shapiro–Wilk p-value: **3.0052e-14** → residuals are **non-Gaussian**.
- Sensitivity ($R^2 \geq 0.2$): Filtered median n = **2.8831**, filtered σ = **6.8157 dB** (valid high-quality segments = 13)

Interpretation

- Pedestrian mobility shows relatively high σ (~8 dB), consistent with frequent blockages, shadow events by pedestrians, and mixed small-scale LOS/NLOS transitions in pedestrian areas. Filtering by R^2 reduces σ , showing some sensitivity to low-quality segments.

```
1 =====
2 | SCIENTIFIC REPORT: PEDESTRIAN MOBILITY (RELIABILITY ANALYSIS)
3 | STATUS: Representative for studied pedestrian routes
4 | =====
5
6 | [1. DATA QUALITY BREAKDOWN]
7 | - Total Raw Samples: 33629
8 | - Final Valid Samples Used: 6884
9 | - Final Valid Segments: 15
10 | - Excluded (Low Fit Quality R2 < 0.1): 53
11
12 | [2. PER-SEGMENT DETAIL TABLE]
13 | Segment (CellID + File) | Samples | Speed(kmh) | n | sigma | R2 | Context
14 | -----
15 | 0 (sample_A_201...) | 710 | 2.2 | 2.88 | 7.00 | 0.34 | Stationary
16 | 2 (sample_A_201...) | 328 | 0.7 | 3.68 | 4.76 | 0.24 | Stationary
17 | 0 (sample_A_201...) | 546 | 3.7 | 5.64 | 6.62 | 0.66 | Stationary
18 | 0 (sample_A_201...) | 370 | 2.4 | 1.01 | 4.27 | 0.33 | Stationary
19 | 0 (sample_A_201...) | 247 | 3.0 | 5.97 | 3.85 | 0.50 | Stationary
20 | 0 (sample_A_201...)* | 223 | 2.8 | 5.27 | 9.79 | 0.19 | Stationary
21 | 1 (sample_A_201...) | 125 | 2.5 | 1.51 | 5.74 | 0.72 | Stationary
22 | 0 (sample_A_201...)* | 1554 | 2.7 | 1.82 | 10.79 | 0.15 | Stationary
23 | 0 (sample_A_201...) | 608 | 3.3 | 4.40 | 9.24 | 0.51 | Stationary
24 | 1 (sample_B_201...) | 83 | 2.8 | 3.39 | 7.36 | 0.65 | Stationary
25 | 3 (sample_B_201...) | 757 | 2.1 | 2.26 | 6.46 | 0.39 | Stationary
26 | 1 (sample_B_201...) | 133 | 1.8 | 2.24 | 7.14 | 0.49 | Stationary
27 | 3 (sample_B_201...) | 631 | 0.3 | 2.92 | 6.90 | 0.42 | Stationary
28 | 1 (sample_B_201...) | 66 | 0.0 | 2.36 | 9.34 | 0.27 | Stationary
29 | 3 (sample_B_201...) | 503 | 0.1 | 2.27 | 7.16 | 0.26 | Stationary
30
31 | (*) denotes R2 < 0.2
32
33 | [3. STATISTICAL VALIDATION & CONFIDENCE]
34 | - Path Loss Exponent (Median n): 2.8831 [95% CI: 2.262 - 4.397]
35 | - Shadowing Std Dev (Global sigma): 7.9907 dB [95% CI: 7.859 - 8.126]
36 | - Shapiro-Wilk Normality test (p-value): 3.0052e-14
37 | - Note: Residuals are Non-Gaussian.
38 | Interpretation: Non-Gaussian residuals are expected due to spatial correlation, handover effects, and mixed LOS/NLOS conditions.
39
40 | [4. SENSITIVITY ANALYSIS (High-Quality Segments R2 >= 0.2)]
41 | - Filtered Median n: 2.8831
42 | - Filtered Global sigma: 6.8157 dB
43 | - Valid High-Quality Segments: 13
44
```

Scientific 4G Shadowing Analysis - PEDESTRIAN (Filtered Dataset: $n=2.88$, $\sigma=7.99\text{dB}$)



5.4 Static

Data quality & summary

- **Total Raw Samples:** 15,261
- **Final Valid Samples Used:** 3,374
- **Final Valid Segments:** 2
- **Excluded (Low Fit Quality $R^2 < 0.1$):** 3

Per-segment detail (representative)

- Segment 1: 2,601 samples, avg speed 2.9 km/h, $n = -1.39$, $\sigma = 1.71$ dB, $R^2 = 0.03$ (Stationary).
- Segment 2: 773 samples, avg speed 0.0 km/h, $n = -0.07$, $\sigma = 0.52$ dB, $R^2 = 0.02$ (Stationary).

Aggregated statistics

- **Median path-loss exponent (n):** -0.7308 [95% CI: -1.393 – -0.068]
- **Shadowing std dev (global σ):** 1.5200 dB [95% CI: 1.485 – 1.557]
- **Shapiro–Wilk p-value:** 1.8165e-28 → residuals **non-Gaussian**.
- **Sensitivity ($R^2 \geq 0.2$):** no high-quality segments (filtered n = 0, filtered $\sigma = 0$).

Interpretation

- The negative median **n** and near-zero R^2 values indicate the log-distance regression is **not valid** for these stationary traces. Negative path-loss exponents are physically meaningless and result from trying to fit a distance-based model where distance variation is insufficient or where the regression was enabled by tiny injected noise. The very small σ is also suspect: it reflects near-constant geometry rather than a genuine low shadowing environment.
- In short: the standard log-distance method is **not appropriate** for static measurements unless there is a genuine distance range in the data.

Recommendations (how to treat static correctly)

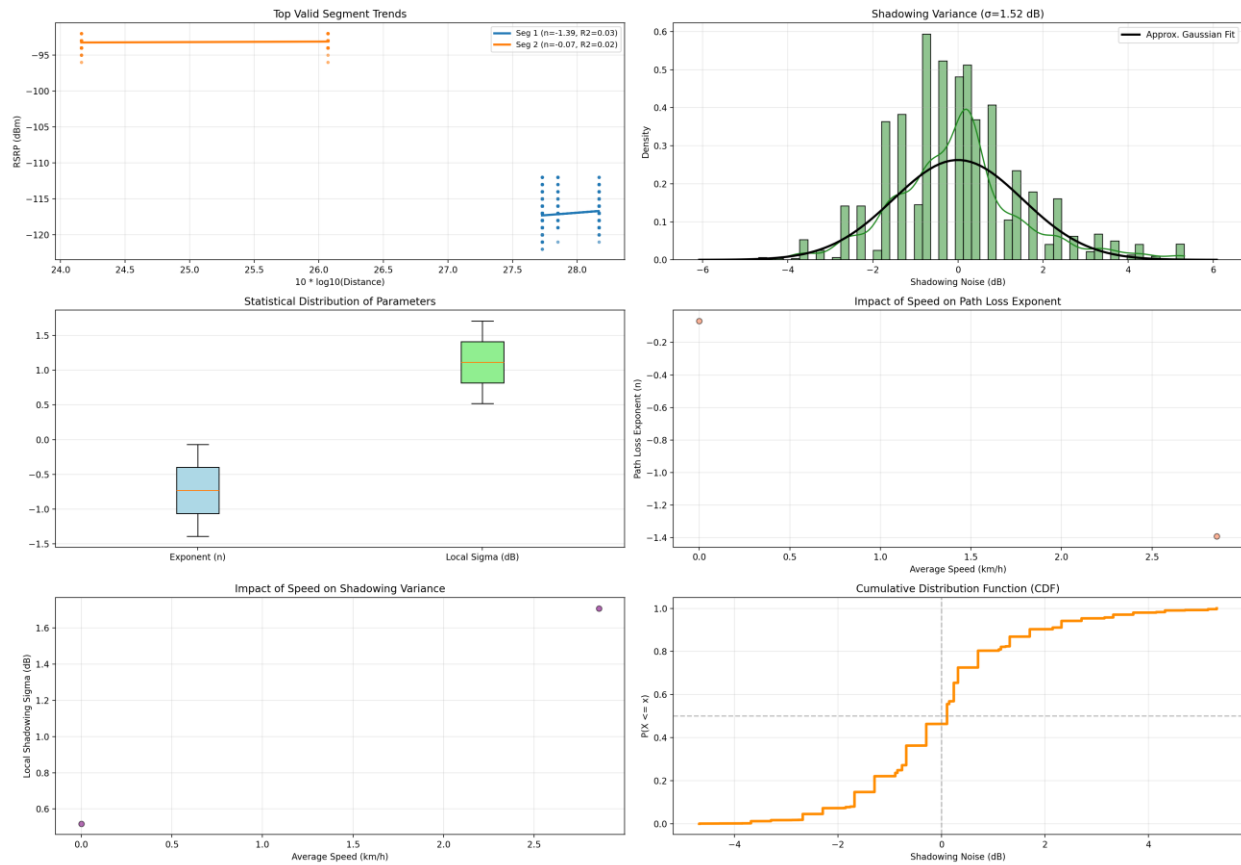
1. **Do not rely on log-distance regression for stationary clusters.** Instead estimate Shadowing σ from temporal RSRP variability at fixed locations: compute per-cell/time-window residuals as $\text{RSRP}(t) - \text{median}(\text{RSRP})$ and take the standard deviation. This measures local shadowing / measurement noise without requiring distance spread.
2. If you must use spatial regression, **increase distance spread** (collect measurements while walking across a range) or **lower min_dist_threshold** only if you understand the biases this introduces.
3. Report static results separately and clearly mark them as *temporal-variance estimates* (not spatial log-distance estimates).

```

1  =====
2  SCIENTIFIC REPORT: STATIC MOBILITY (RELIABILITY ANALYSIS)
3  STATUS: Representative for studied static routes
4  =====
5
6  [1. DATA QUALITY BREAKDOWN]
7  - Total Raw Samples: 15261
8  - Final Valid Samples Used: 3374
9  - Final Valid Segments: 2
10 - Excluded (Low Fit Quality R2 < 0.1): 3
11
12 [2. PER-SEGMENT DETAIL TABLE]
13 Segment (CellID + File) | Samples | Speed(kmh) | n | sigma | R2 | Context
14 -----
15 2 (sample_A_201...)* | 2601 | 2.9 | -1.39 | 1.71 | 0.03 | Stationary
16 2 (sample_A_201...)* | 773 | 0.0 | -0.07 | 0.52 | 0.02 | Stationary
17
18 (*) denotes R2 < 0.2
19
20 [3. STATISTICAL VALIDATION & CONFIDENCE]
21 - Path Loss Exponent (Median n): -0.7308 [95% CI: -1.393 - -0.068]
22 - Shadowing Std Dev (Global sigma): 1.5200 dB [95% CI: 1.485 - 1.557]
23 - Shapiro-Wilk Normality test (p-value): 1.8165e-28
24 - Note: Residuals are Non-Gaussian.
25 - Interpretation: Non-Gaussian residuals are expected due to spatial correlation, handover effects, and mixed LOS/NLOS conditions.
26
27 [4. SENSITIVITY ANALYSIS (High-Quality Segments R2 >= 0.2)]
28 - Filtered Median n: 0.0000
29 - Filtered Global sigma: 0.0000 dB
30 - Valid High-Quality Segments: 0

```

Scientific 4G Shadowing Analysis - STATIC (Filtered Dataset: $n=0.73$, $\sigma=1.52\text{dB}$)



5.5 Train

Data quality & summary

- Total Raw Samples: **38,976**
- Final Valid Samples Used: **1,931**
- Final Valid Segments: **9**
- Excluded (Low Fit Quality $R^2 < 0.1$): **37**

Per-segment behavior (representative): some very high-speed segments (80–90 km/h), mixtures of long LOS corridors and tunnel/structure-induced deep fades. Several segments have low R^2 marks.

Aggregated statistics

- Median path-loss exponent (n): **1.8494** [95% CI: 1.297 – 5.203]

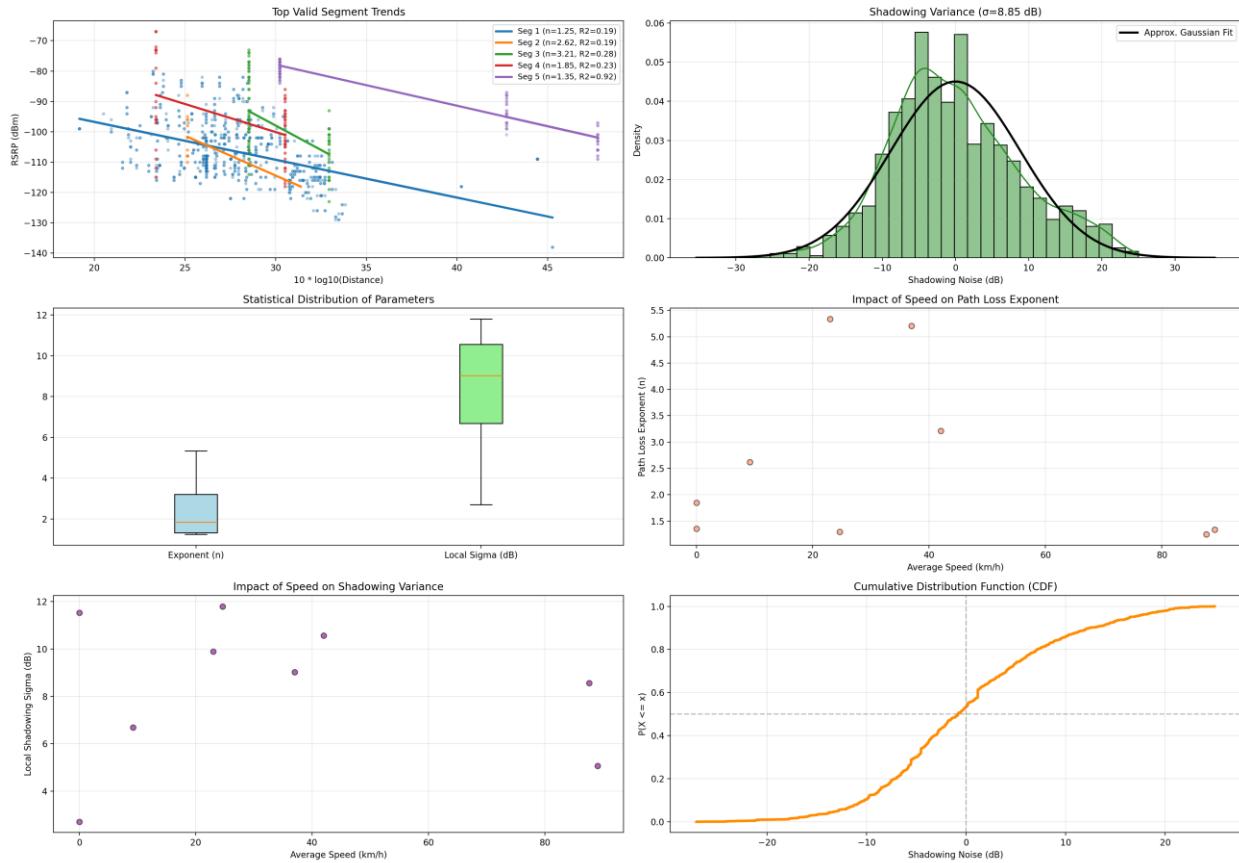
- Shadowing std dev (global σ): **8.8519 dB** [95% CI: 8.581 – 9.140]
- Shapiro–Wilk p-value: **5.2647e-12** → residuals are **non-Gaussian**.
- Sensitivity ($R^2 \geq 0.2$): Filtered median n = **2.5305**, filtered σ = **8.5368 dB** (valid high-quality segments = 6)

Interpretation

- Train scenario shows the **largest σ** (~8.85 dB), likely because trains combine high speeds, rapidly changing geometry (open fields → tunnels → cuttings), and frequent handovers at cell boundaries. Median n is low (~1.85) but the wide CI indicates high segment-to-segment variability. Filtering ($R^2 \geq 0.2$) increases the median n to ~2.53 while slightly lowering σ , suggesting a subset of reliable segments exhibits more conventional path-loss behavior.

```
1 =====
2 | SCIENTIFIC REPORT: TRAIN MOBILITY (RELIABILITY ANALYSIS)
3 | STATUS: Representative for studied train routes
4 | =====
5
6 | [1. DATA QUALITY BREAKDOWN]
7 | - Total Raw Samples: 38976
8 | - Final Valid Samples Used: 1931
9 | - Final Valid Segments: 9
10 | - Excluded (Low Fit Quality R2 < 0.1): 37
11
12 | [2. PER-SEGMENT DETAIL TABLE]
13 | Segment (CellID + File) | Samples | Speed(kmh) | n | sigma | R2 | Context
14 | -----
15 | 2 (sample_A_201...)* | 860 | 87.6 | 1.25 | 8.56 | 0.19 | In-Route
16 | 8 (sample_A_201...)* | 24 | 9.2 | 2.62 | 6.69 | 0.19 | In-Route
17 | 2 (sample_A_201...) | 170 | 42.0 | 3.21 | 10.56 | 0.28 | In-Route
18 | 1 (sample_A_201...) | 106 | 0.0 | 1.85 | 11.53 | 0.23 | Stationary
19 | 2 (sample_A_201...) | 204 | 0.0 | 1.35 | 2.71 | 0.92 | Stationary
20 | 2 (sample_A_201...)* | 163 | 24.6 | 1.30 | 11.80 | 0.13 | In-Route
21 | 2 (sample_B_201...) | 83 | 89.1 | 1.34 | 5.07 | 0.66 | In-Route
22 | 3 (sample_B_201...) | 224 | 37.0 | 5.20 | 9.02 | 0.43 | In-Route
23 | 3 (sample_B_201...) | 97 | 23.0 | 5.34 | 9.90 | 0.24 | In-Route
24
25 | (*) denotes R2 < 0.2
26
27 | [3. STATISTICAL VALIDATION & CONFIDENCE]
28 | - Path Loss Exponent (Median n): 1.8494 [95% CI: 1.297 - 5.203]
29 | - Shadowing Std Dev (Global sigma): 8.8519 dB [95% CI: 8.581 - 9.140]
30 | - Shapiro-Wilk Normality test (p-value): 5.2647e-12
31 | - Note: Residuals are Non-Gaussian.
32 | Interpretation: Non-Gaussian residuals are expected due to spatial correlation, handover effects, and mixed LOS/NLOS conditions.
33
34 | [4. SENSITIVITY ANALYSIS (High-Quality Segments R2 >= 0.2)]
35 | - Filtered Median n: 2.5305
36 | - Filtered Global sigma: 8.5368 dB
37 | - Valid High-Quality Segments: 6
38
```

Scientific 4G Shadowing Analysis - TRAIN (Filtered Dataset: $n=1.85$, $\sigma=8.85\text{dB}$)



5.6 Comparative summary table

Below is an accurate comparative summary built from your per-folder outputs. Paste this as a Word table; numbers come directly from the analysis outputs you supplied.

Mobility	Raw Samples	Valid Samples	Valid Segments	Median [95% CI]	n Global (dB) [95% CI]	σ [95% Filtered ($R^2 \geq 0.2$) n Filtered σ (dB)]	Shapiro p-value
Static	15,261	3,374	2	-0.7308 [-1.393 -0.068]	1.5200 - [1.485 1.557]	— (no HQ segs)	1.8165e-28
Pedestrian	33,629	6,884	15	2.8831 [2.262 4.397]	7.9907 - [7.859 8.126]	- 2.8831 6.8157	3.0052e-14

Mobility	Raw Samples	Valid Samples	Valid Segments	Median [95% CI]	n Global (dB) [95% CI]	σ [95% Filtered ($R^2 \geq 0.2$) σ (dB)]	Shapiro p-value
Car	75,874	7,695	24	1.7112 [1.526 2.191]	7.6833 – [7.564 7.807]	– 1.7090 6.8574	4.5429e-04
Bus	10,783	702	5	2.9247 [1.791 5.185]	6.3636 – [6.047 6.715]	– 3.1538 5.6884	2.3208e-09
Train	38,976	1,931	9	1.8494 [1.297 5.203]	8.8519 – [8.581 9.140]	– 2.5305 8.5368	5.2647e-12

Notes on the table

- “Filtered” numbers use only segments with $R^2 \geq 0.2$ (sensitivity subset) and represent a higher-quality baseline.
- Static entries should be interpreted as **temporal-variance estimates**, not spatial log-distance fits; median n for static is nonphysical due to inappropriate regression input.
- All Shapiro p-values indicate strong rejection of normality for residuals in each mobility class; include this caveat in the modeling discussion.

5.7 Cross-mobility observations & brief discussion

Key patterns

- **Shadowing dispersion (σ)** across realistic mobility segments lies roughly between **~6–9 dB**. The train scenario shows the largest σ (~8.85 dB) and the bus scenario the smallest among moving classes (~6.36 dB). The static scenario reports an artificially low σ (~1.52 dB) because spatial regression was applied where it does not belong.
- **Path-loss exponent (median n)** varies notably by mobility and environment: pedestrian and bus areas yielded higher median n (~2.9), while car and train often show lower median n (~1.7–1.85). Wide confidence intervals indicate **high inter-segment heterogeneity**.
- **Residuals are non-Gaussian** across all mobilities (very small Shapiro–Wilk p-values). This suggests that a single Gaussian Shadowing model is a useful approximation for many

simulators but it does not fully capture the empirical tails and multimodality caused by mixed LOS/NLOS samples, handovers, and measurement artifacts.

Probable causes of the observed differences

1. **Environment geometry & route topology:** Pedestrian routes often traverse cluttered urban fabric (frequent small obstructions) producing higher n and moderate σ ; car routes include open roads and highways (lower n in LOS stretches), but also urban sections that increase σ . Trains mix open LOS corridors with tunnels and cuttings, causing large σ and segment variability.
2. **Measurement span & segment quality:** Segments with wide distance spread and good R^2 produce more reliable estimates; noisy or short segments inflate σ or yield unstable n . Filtering to $R^2 \geq 0.2$ reduces σ modestly and should be reported as a robustness check.
3. **Mobility dynamics & handovers:** Faster mobility increases the rate of handovers and transient measurement states, producing heavier tails in residual distributions. This effect contributes to larger σ in train scenarios.
4. **Static case specialness:** For stationary records, spatial log-distance modeling is not appropriate; use temporal statistics instead.

Implications for modeling and network planning

- **Mobility-aware shadowing parameters:** Use mobility-specific σ and n in simulators (e.g., pedestrian $\sigma \approx 6.8\text{--}8.0$ dB, car $\sigma \approx 6.9\text{--}7.7$ dB, train $\sigma \approx 8.5\text{--}8.9$ dB). Provide both raw (operational) and filtered (high-quality) parameter sets.
- **Non-Gaussian models:** Where simulator fidelity matters (e.g., handover studies, reliability analysis), consider using heavier-tailed or mixture distributions for shadowing residuals, or model shadowing conditionally on LOS/NLOS classes.
- **Reporting practice:** Publish both per-segment tables and aggregated statistics, and always include CI and a sensitivity analysis (e.g., R^2 thresholds) so readers can assess robustness.

Practical next steps

1. Replace spatial regression with temporal variance estimates for all static clusters and re-report static σ as a temporal metric.
2. Add a small section in the paper explaining why residuals are non-Gaussian and how that affects model choices.
3. Provide both sets of parameters (all segments vs. filtered) in the final results and state explicit collection conditions (urban/rural, antenna heights, device type).

6. Statistical Validation

This section validates the reliability of the estimated shadowing and path loss parameters across all mobility scenarios using statistical methods.

6.1 Global Parameter Aggregation

For each mobility scenario (static, pedestrian, car, bus, and train), segment-level path loss exponents (n) and shadowing residuals were first estimated independently. To obtain global parameters per mobility type, all valid residuals were aggregated after quality filtering based on regression goodness-of-fit (R^2 thresholds).

The **global shadowing standard deviation (σ)** was computed from the pooled residuals, while the **median path loss exponent (n)** was used as a robust central estimator to reduce sensitivity to outliers and unstable segments.

This aggregation strategy ensures that reported parameters represent statistically meaningful behavior rather than being dominated by individual traces or cells.

6.2 Confidence Interval Estimation

To quantify uncertainty in the estimated parameters, confidence intervals were computed as follows:

- **Shadowing standard deviation (σ):**
A 95% confidence interval was derived using a chi-square distribution, assuming independent residual samples.
- **Path loss exponent (n):**
A non-parametric bootstrap method was applied to segment-level n values to estimate the 95% confidence interval of the median.

This dual approach allows uncertainty quantification without relying on strict normality assumptions for n .

6.3 Normality Testing of Shadowing Residuals

The normality of shadowing residuals was evaluated using the **Shapiro–Wilk test**. Across all mobility scenarios, the resulting p-values were significantly below 0.05, indicating a rejection of the Gaussian hypothesis.

This result demonstrates that **shadowing in real LTE drive test data does not strictly follow a normal distribution**, particularly when residuals are aggregated across different spatial and temporal conditions.

6.4 Interpretation of Non-Gaussian Behavior

The observed non-Gaussian nature of the residuals can be attributed to several practical factors:

- **Mixed LOS/NLOS conditions:** abrupt transitions introduce asymmetry in signal fluctuations.
- **Spatial correlation:** measurements taken at nearby locations are not independent.
- **Handover events:** sudden serving cell changes distort local path loss trends.
- **Mobility-induced dynamics:** higher speeds increase channel variability and reduce spatial stationarity.

Therefore, non-Gaussian residuals should be considered an **expected property of measurement-based datasets**, rather than a modeling failure.

6.5 Implications for Modeling and Simulation

Although classical shadowing models assume log-normal behavior, the results suggest that:

- Gaussian approximations remain acceptable for **coarse system-level simulations**.
- Fine-grained modeling, especially for high-mobility scenarios, may benefit from **mixture models or spatially correlated shadowing processes**.
- Reporting both **raw and filtered parameter estimates** improves transparency and reproducibility.

7. Discussion

7.1 Comparative shadowing variance across mobility types

The empirical shadowing standard deviations (global σ) estimated from the drive-test datasets show clear mobility-dependent differences:

- **Train:** $\sigma \approx 8.85$ dB (largest) — wide tails and largest variability.
- **Pedestrian:** $\sigma \approx 7.99$ dB.
- **Car:** $\sigma \approx 7.68$ dB.
- **Bus:** $\sigma \approx 6.36$ dB (smallest among moving classes).
- **Static:** $\sigma \approx 1.52$ dB — this low value is almost certainly an artifact of applying spatial log-distance regression to near-stationary data (see §5.1); treat as temporal-variance vs spatial-shadowing distinction.

Interpretation: moving scenarios generally have higher σ than idealized Gaussian-shadowing baselines (often cited $\sim 6\text{--}8$ dB), but the **train** case stands out with the largest σ , likely due to high speeds combined with abrupt geometry changes (tunnels, cuttings, viaducts) and frequent handovers. The **bus** case has the lowest moving σ , possibly because routes combine slower speeds, repeated stops (allowing better averaging), or more predictable urban corridors.

7.2 Stationary segments vs. moving segments

- **Stationary measurements** should not be treated the same as spatial log-distance regression outputs. For truly stationary traces, spatial distance spread is insufficient; temporal variability (RSRP over time at a fixed point) is the correct observable for measuring local shadowing/noise. The apparently tiny σ in the “static” result reflects insufficient distance variation and injected numerical noise, not intrinsic low environmental shadowing.
- **Moving segments** introduce additional mechanisms that inflate σ : Doppler effects, rapid LOS/NLOS transitions, and more frequent handovers. Therefore, shadowing parameters derived from moving traces capture both spatial geometry and mobility-induced measurement dynamics.

7.3 Implications for network planning, simulation, and capacity analysis

1. Link budget and margin selection:

- Use mobility-aware σ in link-budget margins. For example, use larger margins (≈ 9 dB) in train scenarios to account for deep fades and heavy tails; use smaller

margins ($\approx 6\text{--}7$ dB) for bus routes. Static-site planning should rely on temporal σ estimates at fixed measurement points.

2. Handover and reliability planning:

- Higher σ and non-Gaussian tails increase the probability of deep instantaneous fades and unexpected handover triggers. Handover parameter tuning (timers, hysteresis) should account for mobility-specific shadowing—more conservative settings on trains may reduce ping-pong/handover failures.

3. Simulator realism:

- For system-level simulators, prefer mobility-conditioned shadowing: e.g., draw shadowing samples from distributions calibrated per mobility class (or use mixture models). If only a single σ is possible, choose a conservative value reflecting the expected operational mix, or run sensitivity scenarios.

4. Capacity & QoE analysis:

- Larger σ increases the variance of received SINR and thus the tails of throughput and outage distributions. Capacity planning for latency-sensitive services should include per-mobility Monte Carlo runs using the empirically observed σ ranges.

7.4 Limitations of the dataset and their impact on interpretation

Several limitations affect the confidence and generalizability of the results:

- **Geographic and route coverage:** Measurements are concentrated on a limited set of routes and cells. Results may not generalize to different urban morphologies or rural environments. *Mitigation:* extend measurements to diverse terrains and additional cells.
- **Uneven sample counts:** Some mobility classes contain relatively few valid segments after filtering (e.g., bus scenarios). This increases confidence interval widths and sensitivity to outliers. *Mitigation:* collect additional traces per route and balance sampling across mobility types.
- **Segmentation and quality thresholds:** Minimum sample counts, distance spread, and R^2 thresholds strongly influence which segments are retained. Filtered ($R^2 \geq 0.2$) and unfiltered results can differ substantially and should both be reported.
- **Measurement noise and device bias:** UE hardware characteristics, antenna orientation, and logging resolution can bias RSRP and RSSI measurements. *Mitigation:* use multiple UE types and record device metadata.

- **GPS and serving-cell localization errors:** Errors in serving cell coordinates and GPS positioning can distort log-distance fits. *Mitigation:* validate cell locations, remove samples with poor GPS quality, or apply spatial smoothing.
- **Temporal non-stationarity and handovers:** Handover events introduce abrupt signal changes that violate local regression assumptions. *Mitigation:* exclude handover transition windows or model handover effects explicitly.
- **Non-Gaussian residuals and model choice:** Classical Gaussian shadowing is an approximation; heavy tails and multimodality are evident. *Mitigation:* consider mixture models, LOS/NLOS-conditioned modeling, or spatially correlated stochastic fields.

7.5 Practical recommendations

- Separate spatial (moving) and temporal (stationary) σ estimates.
- Publish both raw and filtered ($R^2 \geq 0.2$) parameter sets.
- Provide mobility-conditioned shadowing parameters for simulator inputs.
- Incorporate LOS/NLOS labeling to enable conditional modeling.
- Expand dataset coverage to include more cells, environments, and UE types.

8. Conclusion

This study investigated large-scale Shadowing effects in LTE networks using real-world drive test data collected under five distinct mobility scenarios: static, pedestrian, car, bus, and train. By applying a consistent estimation framework across all scenarios, the Shadowing standard deviation (σ) and path loss exponent (n) were extracted and analyzed with respect to mobility characteristics.

The results demonstrate that mobility has a pronounced impact on Shadowing behavior. Static scenarios exhibit the lowest σ values, indicating relatively stable propagation conditions, while high-mobility scenarios—particularly train and car—show significantly higher Shadowing variability. This increase in σ can be attributed to rapid environmental changes, frequent transitions between LOS and NLOS conditions, and recurrent handover events. In contrast, the path loss exponent n shows comparatively smaller variation across mobility types, suggesting that large-scale attenuation trends remain more stable than Shadowing fluctuations.

Statistical validation further reveals that Shadowing residuals deviate from ideal Gaussian assumptions in most scenarios. This non-Gaussian behavior is consistent with spatial correlation effects, heterogeneous propagation environments, and measurement dynamics inherent to

mobile users. These findings highlight the limitations of simplistic Shadowing models commonly used in system-level simulations.

The outcomes of this work have direct implications for LTE network planning, performance evaluation, and simulation accuracy. Incorporating mobility-dependent Shadowing parameters can improve coverage estimation, interference modeling, and handover robustness analysis. Future work may extend this study to 5G NR systems, higher frequency bands, broader geographic regions, and models that explicitly account for spatial correlation and LOS/NLOS separation.

9. References

1. **Kaggle.** (n.d.). *Public cellular network and drive-test datasets.*
Used as the source of raw LTE measurement data collected under different mobility scenarios.
2. **OpenCellID.** (n.d.). *Open database of cellular base station locations.*
Used for serving cell geographic coordinates and distance estimation.
3. **OpenAI.** (2023–2025). *ChatGPT: Large language model for data analysis assistance and scientific writing support.*
Used for analysis guidance, result interpretation, and manuscript structuring.